

Optimal Tracking and Teacher Discretion

Update — May 2026

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New descriptives

Who follows the recommendation?

Alignment between teacher advice and student track choice (G2 → G3)

	Follow advice	Deviate UP	Deviate DOWN
All students	58%	31%	11%
<i>By SES</i>			
Low SES	58%	29%	13%
High SES	56%	37%	7%
<i>By gender</i>			
Female	60%	35%	6%
Male	55%	27%	18%
<i>By cycle architecture</i>			
C1 (cycle-1 only)	60%	29%	11%
C123 (full cycle)	55%	32%	12%

$N = 4,438$. “UP” = student chooses a higher track than advised (e.g. advised TSO, chooses ASO). “DOWN” = opposite.

Summarized:

- **Schools recommend conservatively.** Deviations are 3:1 upward vs. downward. Students are systematically more ambitious than schools advise.
- **High-SES students override upward more (37% vs. 29%) and deviate down less (7% vs. 13%).** They have the confidence and information to correct conservative advice.
- **Low-SES students comply more with downward advice.** The recommendation disproportionately binds for low-SES students — not through differential treatment, but through differential compliance.
- **C123 schools are more conservative than C1.** Lower compliance (55% vs. 60%), more upward deviation (32% vs. 29%). Consistent with retention incentives generating strategic conservatism.
- **Boys deviate down much more than girls (18% vs. 6%)** — likely choosing vocational/technical paths even when school advises academic.
- **Note:** it can also be that some groups receive more conservative (or biased) advice on average (i.e women not being recommended to stem all else equal)

Student's choice problem

- $t = 0$ (**Baseline**): Initial measurements taken and school enrollment realized.
- $t \in \{1, 2\}$ (**Cycle 1**): End-of-year standardized exams administered and school recommendations delivered.
- $t = 3$ (**Track-subtrack Choice**): Binding initial track choice made between ASO, TSO, and BSO and stem nonstem.
- $t \geq 4$ (**Cycles 2 & 3**): Potentially repeat 3 for each cycle 2-3 for each cycle.

Sources of heterogeneity

1. **Known Type (θ_i^k):** Discrete profile capturing cognitive, verbal, non-cognitive, and family traits. Known to agents but latent to us; identified via baseline measurements.
2. **Unknown Match Quality (θ_i^u):** A 5-dimensional vector of track-specific match qualities. Unknown at $t = 0$ and learned over time.
3. **Observed Covariates (X_{it}):** Exogenous controls like gender, age, and grade level.

Production function

Standardized grades G_{ibt} in Math and Dutch (b) provide public signals about θ_i^u :

$$G_{ibt} = \mu_b(\theta_i^k, X_{it}) + \Lambda_b^\top \theta_i^u + \epsilon_{ibt}$$

- ϵ_{ibt} is independent Gaussian noise.
- $\Lambda_b^\top$ maps test subjects to track and STEM/Non-STEM comparative advantages.
- **Key advantage:** The balanced panel of standardized exams is observed for all students regardless of track, allowing selection-free identification of learning parameters.

Students update their beliefs about θ_i^u sequentially within each period:

- **Stage 1 (Grades):** Standard Bayesian updating yields a closed-form Gaussian posterior.
- **Stage 2 (Recommendations):** given grades, the discrete school advice forces an update via an ordered probit likelihood, creating a mixture-of-truncated-normals. Solved via Density Filtering (in progress)
- **Intuition:** recommendations are informative beyond grades, as we also observe in the data. Can shift up or down the beliefs of the student's fit for each track-subtrack.

Students maximize expected lifetime utility over track paths, where flow utility depends directly on their subjective belief means (which advice affects).

- **Standard DDC:** Standard forward-looking maximization over available tracking actions.
- **Finite Dependence/absorbing state:** we can use dropout or BSO as absorbing state to simplify CCP inversion.
- **When is the last period?** Interested in HE too or not?

School's recommendation problem

Two-stage decomposition

Stage 1: Track recommendation (capacity-constrained).

- School constructs academic index per track: $I_{ij}^{\text{track}} = W_{it}^{\top} \alpha_{j,\tau} + \nu_{ijt}$
- Ranks students, recommends top $x_j^s = (1 + b_s) \cdot K_{jt}^s / N_s$ to each track
- Vertical ordering: ASO > TSO > BSO on academic ability
- Estimated how? $\max_j \{I_{ij}^{\text{track}} - \kappa_j\}$ where κ_j are implicit cutoffs.

Stage 2: Subtrack recommendation (unconstrained).

- Conditional on track, school assesses STEM vs. nonSTEM comparative advantage
- $R_i^{\text{sub}} = \text{STEM} \iff W_{it}^{\top} \gamma_{S,\tau} > W_{it}^{\top} \gamma_{N,\tau} + \omega$
- Key covariates: relative math rank vs. relative Dutch rank (Z rankings)
- No capacity constraint — estimated as binary logit within track

Stackelberg structure

The school doesn't pick $\alpha_{j,\tau}$ arbitrarily — it anticipates student responses.

School's per-period problem:

$$\max_{\alpha_\tau, \gamma_\tau} \sum_i \sum_j P(d_{it} = j \mid R_{it}(\alpha, \gamma), s_{it}) \cdot \pi_{ij\tau}$$

subject to track-level capacity: $\sum_i P(R_i^{\text{track}} = j) = x_j^s$

Components:

- $P(d_{it} = j \mid R_{it}, s_{it})$: student's choice probability from estimated dynamic model (beliefs updated via grades + recommendation)
- $\pi_{ij\tau}$: school's payoff = $E^{\text{school}}[G_{ij,t+1}] + \omega_\tau \cdot P(\text{pass})$.
- ω_τ : weight on pass rates. $\omega_{C123} > \omega_{C1}$ generates conservative recommendations at C123 schools

Fixed point: changing $\alpha \rightarrow$ changes $R \rightarrow$ changes beliefs (moment-matching) $\rightarrow$ changes $P(d = j) \rightarrow$ school must anticipate this chain.

The core tension

The recommendation is **simultaneously informative and distorted**:

Information value

- School observes Z (relative rank, effort) — private info students lack
- Recommendation helps students match to right track
- Especially valuable for low-SES students with weak priors

Strategic distortion

- C123 schools push marginal students down to protect pass rates and manage capacity
- High-SES students override $\Rightarrow$ unharmed
- Low-SES students comply $\Rightarrow$ locked into lower tracks
- But: some pushed-down students would have *failed* in ASO — distortion also protects

Net effect is ambiguous

Questions for discussion

1. **Two-stage decomposition:** is it right that capacity binds at the track level only? At least that is the only element we observe in our dataset.
2. **Embedding the A/B/C certificate** (pending checking the data with capacity if mechanical, if not → add another step)
3. **It is feasible to impose Stackelberg structure?** the fixed point requires iterating between the school's rule and student responses (R_{it} affects beliefs, CCPs etc)...student's problem in this case is dynamic.
4. **School's objective:** is next-period grades + pass rate bonus the right specification? Or do we need longer-run outcomes (HE entry, retention)?
5. **Production function / potential outcomes:** balanced grade panel identifies θ^u without selection correction. Is this sufficient for the welfare evaluation, or do we need HE outcomes too?